

MATLAB

Drone Payload Capacity and Structural Design Analysis

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PROGRAM

Engineering Pathways Program, MathWorks Classroom Challenge Projects

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Abstract

Modern quadcopter drones must balance payload capacity, structural safety, and material cost within a fixed thrust budget. Every gram added to the arm structure directly reduces the payload the drone is able to lift, while every material choice changes how much stress and displacement the arm experiences under motor loading. This project developed a MATLAB based design and analysis workflow to evaluate two distinct drone arm geometries across six candidate materials, using thrust to weight analysis, finite element analysis, and material cost optimization to identify the combination that best satisfies the required payload, safety, and cost constraints.

We proposed two arm designs representing different structural strategies. Design 1 is a tapered hollow rectangular tube that concentrates material near the root, where bending loads are highest, and removes material toward the motor mount, where less structural reinforcement is required. Design 2 is a hollow truss shaped structure intended to reduce mass while adding internal reinforcement through triangulated truss members. Both designs were evaluated using a single arm finite element model under the combined upward motor thrust and downward motor weight loading, across Carbon Fiber Composite (CFRP), Aluminum Alloy, Fiberglass Composite (GFRP), PLA Plastic, ABS Plastic, and Wood (Birch).

Our team specifically compared these two geometries because they represent opposite structural philosophies: material efficiency through tapering versus added stiffness through internal triangulation, and we wanted to determine through quantitative analysis, rather than assumption, which strategy produced the stronger and more practical drone arm. Developing a repeatable MATLAB workflow that evaluates geometry and material together, instead of treating them separately, demonstrates how computational design tools can support real engineering decision making early in a product design cycle. The completed analysis identified Design 1, manufactured from Fiberglass Composite, as the design that provided the best overall balance of payload capacity, structural safety, manufacturability, and material cost.

a. Problem Definition and Proposed Solution

Problem Statement

The objective of this project is to design and evaluate a quadcopter drone arm that maximizes payload capacity while maintaining safe flight performance and structural integrity. The drone arm must be evaluated across multiple geometries and multiple material options so that the final recommendation is supported by quantitative results rather than a single untested assumption.

Requirements

- The drone must support a minimum payload of 0.5 kg.
- The drone must meet a thrust to weight ratio of at least 2 to 1 to ensure safe and stable flight.
- The quadcopter configuration uses 4 motors and 4 propellers, with each motor capable of providing up to 1 kg of thrust.
- The total mass of the drone without arms or payload is 1 kg.
- The design must use at least two distinct arm geometries that are meaningfully different from one another.
- Each arm geometry must be evaluated across all six available material options.

- The structural design must maintain an acceptable factor of safety, with a target engineering range of approximately 1.5 to 2 or higher.
- Material cost is considered as an additional optimization criterion once the payload and safety requirements are satisfied.

Dataset and Software

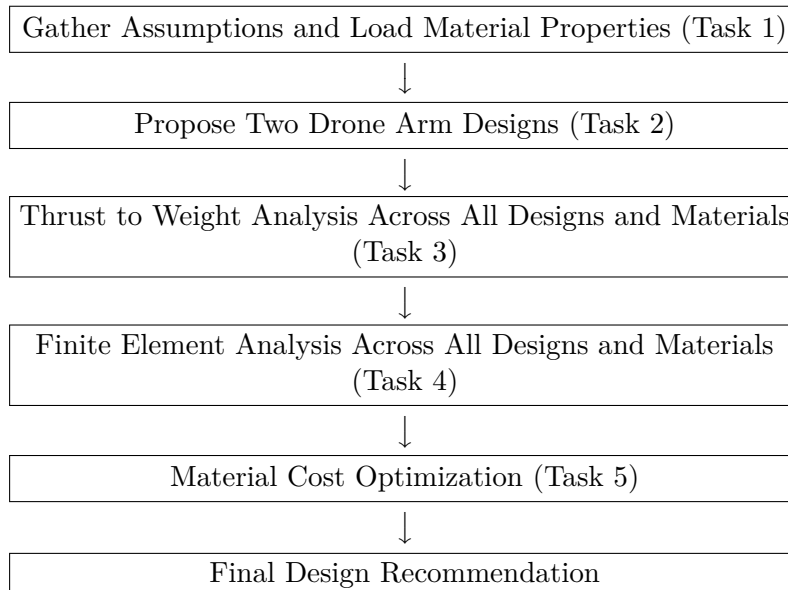
- Software: MATLAB R2026a, Partial Differential Equation Toolbox
- Material Database: `droneArmMaterials.mat`, containing density, Young's modulus, Poisson's ratio, yield strength, and average cost per meter for six candidate materials
- CAD Models: `Design_1_CAD_Model.STEP` and `Design_2_CAD_Model.STL`

Constraints

- The analysis is simplified to a single drone arm rather than the complete four arm drone assembly.
- Material properties are treated as homogeneous and isotropic, which is an approximation for composite and wood materials.
- The arm geometry and loading conditions are idealized rather than representing the complete flight environment.
- The finite element model captures only the dominant motor thrust and motor weight loading, while vibration, landing impact, and aerodynamic forces are not included.

Proposed Solution

Compare two candidate drone arm designs together across six materials using thrust to weight analysis and single arm finite element analysis, then select a final recommendation by comparing payload capacity, structural safety, and material cost side by side.



b. Project File Organization

Table 1: Repository Folder Purpose Summary

Folder	Purpose
CAD-Designs/	CAD models and concept sketches for both drone arm designs
Documentation/	Supporting project documents, including design constraints, assumptions, and the team agreement
MATLAB/	Live Script, plain code export, and the material properties database
Results/	Exported finite element results, payload comparison charts, and material cost tables, organized separately for Design 1 and Design 2

c. Material and Method Reference

This section documents the material properties and the core analysis method used throughout the project, in the same reference style used for the rest of the workflow. The density, yield strength, and cost per meter values were originally reconstructed directly from the project results using the same relationships defined in the analysis script, then confirmed against `Documentation/Drone_Design_Constraints_Assumptions_and_Ideas.xlsx`, the source document used to build `droneArmMaterials.mat`. Every reconstructed value matched the source document exactly. Young’s modulus and Poisson’s ratio are taken directly from that same source document.

Material Properties Reference

Density was recovered from each material’s total arm mass using $\text{TotalArmMass} = \text{ArmVolume} \times \text{Density} \times \text{NumberOfArms}$. Yield strength was recovered from the reported factor of safety and maximum von Mises stress using $\text{FactorOfSafety} = \text{YieldStrength} / \text{MaximumVonMisesStress}$. Cost per meter was recovered from the total material cost using $\text{TotalMaterialCost} = \text{CostPerMeter} \times \text{ArmLength} \times \text{NumberOfArms}$. All three relationships were checked independently against both the Design 1 and Design 2 result sets and produced consistent values in every case, and all three were subsequently confirmed against the source constraints document.

Table 2: Material Properties Reference for the Six Candidate Materials

Material	Density (kg/m ³)	Young’s Modulus (GPa)	Poisson’s Ratio	Yield Strength (MPa)	Cost per Meter (USD/m)
Carbon Fiber Composite (CFRP)	1600	70	0.30	600	30.00
Aluminum Alloy	2700	69	0.33	275	7.50
Fiberglass Composite (GFRP)	1900	25	0.28	300	15.00
PLA Plastic	1250	3.5	0.36	60	3.50
ABS Plastic	1050	2.0	0.35	40	3.50
Wood (Birch)	600	10	0.35	80	2.00

Young’s modulus and Poisson’s ratio are passed directly into the `materialProperties` object used by the finite element solver in Section 4. Their influence is visible in the reported displacement results, since a stiffer material with a higher Young’s modulus produces a smaller displacement under the same load. This trend holds throughout Section 4: CFRP and Aluminum Alloy, the two

stiffest materials at 70 GPa and 69 GPa, produced the smallest displacements on both designs, while PLA and ABS, the two least stiff materials at 3.5 GPa and 2.0 GPa, produced the largest.

thrustToWeightAnalysis

Purpose

Calculates the maximum payload capacity for every combination of drone arm design and material, and determines whether each combination satisfies the minimum payload requirement and the required thrust to weight ratio.

Key Operations

- Loops through each arm design and each material option.
- Calculates the mass of all four drone arms from the arm volume, material density, and number of arms.
- Calculates the maximum payload as the difference between the maximum allowed total drone mass and the combined mass of the drone body and the arms.
- Marks each combination as PASS or FAIL depending on whether it meets both the minimum payload and the thrust to weight requirement.
- Generates a payload comparison bar chart for each design, with passing materials shown in green and failing materials shown in red.

Input

- `armVolumes`, the volume of a single arm for each proposed design
- `designNames`, the label used for each design
- `materials`, the material properties structure loaded from `droneArmMaterials.mat`
- `totalDroneMassWithoutArmsOrPayload` and `totalThrust`, the fixed drone assumptions from Section 1
- `minimumPayload`, `requiredThrustToWeightRatio`, and `numberOfArms`, the project requirements from Section 1

Output

- `task3Results`, a structure containing the design names, material names, maximum payload for every combination, and the pass or fail status used later during material cost optimization

1. Design Assumptions and Material Properties

Task 1

The first stage of the project defines the drone design assumptions, loads the material properties, and establishes the mass budget available for the arms and the payload once the rest of the drone components are accounted for.

1.1 Design Parameters and Constraints

- Quadcopter configuration with 4 motors and 4 propellers
- Maximum thrust of 1 kg per motor, for a total available thrust of 4 kg
- Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$
- Total mass of the drone components without the arms or the payload, 1 kg
- Minimum payload requirement of 0.5 kg
- Minimum thrust to weight ratio of 2 to 1
- Four drone arms

The thrust to weight requirement of 2 to 1 is a system level flight safety standard, distinct from the structural factor of safety evaluated later through finite element analysis. Conceptually, a 2 to 1 thrust to weight ratio means the drone's total available thrust must be at least twice its total weight, which provides margin for stable hovering and maneuvering rather than flying at the very edge of what the motors can lift.

1.2 Component Mass Breakdown

The 1 kg drone mass budget, excluding the arms and payload, is distributed across the following components.

Table 3: Drone Component Mass Breakdown

Component	Quantity	Unit Mass (kg)	Total Mass (kg)
Battery	1	0.30	0.30
Motors	4	0.065	0.26
Propellers	4	0.010	0.04
Electronic Speed Controllers	4	0.030	0.12
Flight Controller and Electronics	1	0.05	0.05
Receiver and GPS	1	0.03	0.03
Drone Frame, Center Body Only	1	0.20	0.20
Total, Excluding Arms and Payload			1.00

Because the quadcopter carries four motors at 1 kg of thrust each, the maximum allowable total mass of the drone, under the required 2 to 1 thrust to weight ratio, is 2 kg. Since 1 kg of that budget is already committed to the components listed above, the remaining 1 kg of budget must be shared between the four drone arms and the payload. This is the central tension the rest of

the analysis is built around: every gram spent on arm material is a gram that cannot be spent on payload.

1.3 Material Properties Overview

Six materials commonly used in drone construction were evaluated for both arm geometries: Carbon Fiber Composite (CFRP), Aluminum Alloy, Fiberglass Composite (GFRP), PLA Plastic, ABS Plastic, and Wood (Birch). The density, yield strength, and cost per meter used for each material are listed in Table c.1 above. These six materials were chosen because they span a wide range of stiffness, strength, weight, and cost, from very strong and expensive composites to inexpensive but lower strength plastics and wood, which allowed the analysis to fairly compare tradeoffs across the full spectrum of realistic drone build materials rather than only comparing similar materials to one another.

2. Drone Arm Design Proposals

Task 2

Two arm designs were developed to represent meaningfully different structural strategies rather than small variations of the same idea. All dimensions for both designs are provided in millimeters for consistency.

2.1 Design 1, Hollow Tapered Rectangular Tube

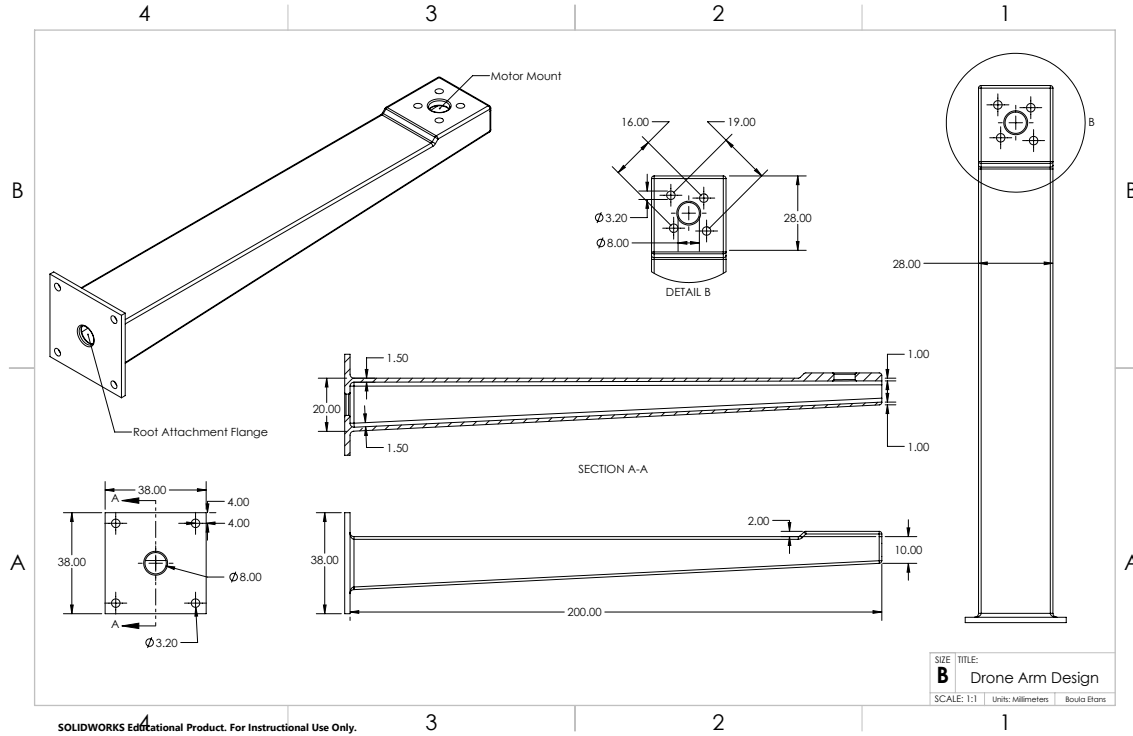


Figure 1: Design 1 Concept Sketch, Hollow Tapered Rectangular Tube

Geometry: A tapered hollow rectangular tube spanning a total length of 200.00 mm. The structure originates at the base with a 38.00 mm by 38.00 mm square attachment flange and a profile measuring 28.00 mm in width and 20.00 mm in height, with a wall thickness of 1.50 mm. The arm tapers toward the distal end to a profile measuring 28.00 mm in width and 10.00 mm in height, with a reduced wall thickness of 1.00 mm. The arm terminates in a 28.00 mm by 28.00 mm square motor mount platform positioned 2.00 mm above the top edge of the tapered body.

Reasoning: This geometry was chosen to optimize the strength to weight ratio. A drone arm acts as a cantilever beam, meaning the highest bending moments occur at the base where it attaches to the frame, while the stresses at the motor end are significantly lower. By tapering both the overall profile and the wall thickness, the design strategically places material only where it is needed to carry the load, removing unnecessary mass at the tip rather than carrying a constant, oversized cross section along the entire arm.

Analysis Prediction: Under the 1 kg upward thrust load, this arm was expected to show strong bending stiffness at the root, keeping overall deflection low. Because the tapered profile minimizes

total arm mass, the design was also expected to consume less of the drone's 2 to 1 thrust to weight budget, leaving more of that budget available to carry the required 0.5 kg payload.

2.2 Design 2, Hollow Truss Structure

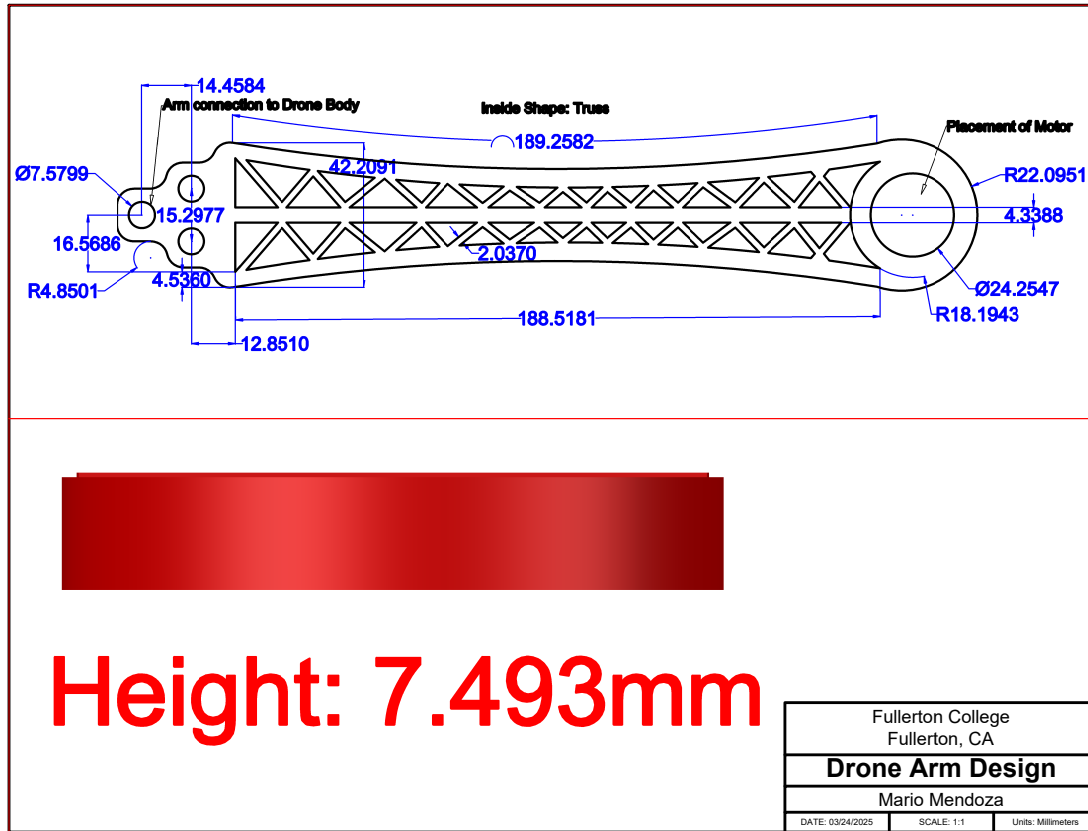


Figure 2: Design 2 Concept Sketch, Hollow Truss Structure

Geometry: A hollow, truss shaped interior structure where each truss branch has a thickness of approximately 2.0370 mm. The outer walls feature a continuous inward curve, with a radius of 189.2582 mm, causing the internal truss structure to narrow toward the middle of the arm before widening again as it approaches the motor housing. The holes used for the arm connection are arranged in a triangular pattern. The total height of the drone arm is 7.493 mm throughout its length.

Reasoning: The truss shaped interior was designed for stabilization, structural efficiency, and weight reduction. This arrangement also leaves room for future flexibility in motor placement, since the connection to the drone body is shaped as a triangle to keep the attachment point secured and geometrically rigid rather than relying on a single flat mounting face.

Analysis Prediction: While the truss interior was expected to provide meaningful strength and stiffness, the transition region between the arm connection point and the start of the curved section was identified as a possible concern before any analysis was run. It was hypothesized that this transition could become a weak point under load, and further optimization of that region might be required if the finite element results showed it to be a limiting factor.

2.3 Comparison of the Proposed Geometries

Table 4: Geometry Comparison Between the Two Proposed Drone Arm Designs

Property	Design 1	Design 2
Structural Strategy	Tapered wall and profile, material concentrated at the root	Internal truss reinforcement, material distributed through triangulated branches
Overall Arm Length	200.00 mm	251.00 mm
Single Arm Volume	$2.483 \times 10^{-5} \text{ m}^3$ (24.83 cm ³)	$3.430 \times 10^{-5} \text{ m}^3$ (34.30 cm ³)
Anticipated Structural Risk	Reduced wall thickness at the tapered tip	Transition region between the connection point and the curved truss section

Before any thrust to weight or finite element analysis was performed, Design 1 already showed a smaller single arm volume than Design 2, approximately 24.83 cm³ compared to 34.30 cm³. Since both designs use the same material set, a smaller volume directly translates into less arm mass for any given material, which gave an early indication that Design 1 would consume less of the drone’s thrust to weight budget before either design was evaluated under load.

3. Thrust to Weight Analysis

Task 3

The purpose of this stage was to evaluate which drone arm design and material combination maximizes payload capacity while meeting the required flight safety standards. With two designs and six materials, the analysis produced twelve payload capacity estimates in total.

3.1 Methodology

The `thrustToWeightAnalysis` function, documented in Section c above, loops through both arm designs and all six materials. For each combination, the function calculates the total mass of all four arms from the arm volume and material density, then subtracts that arm mass and the fixed 1 kg drone component mass from the maximum allowable total drone mass of 2 kg to obtain the maximum payload the design can carry while still meeting the 2 to 1 thrust to weight ratio. A combination is marked as PASS only if the resulting maximum payload is at least 0.5 kg and the thrust to weight ratio at that payload is still at least 2 to 1.

3.2 Design 1 Results

Table 5: Design 1 Maximum Payload Results

Material	Total Arm Mass (kg)	Maximum Payload (kg)
Carbon Fiber Composite (CFRP)	0.15893	0.84107
Aluminum Alloy	0.26819	0.73181
Fiberglass Composite (GFRP)	0.18872	0.81128
PLA Plastic	0.12416	0.87584
ABS Plastic	0.10429	0.89571
Wood (Birch)	0.059597	0.94040

Design 1 passed the payload and thrust to weight requirement for all six materials. No material combination failed.

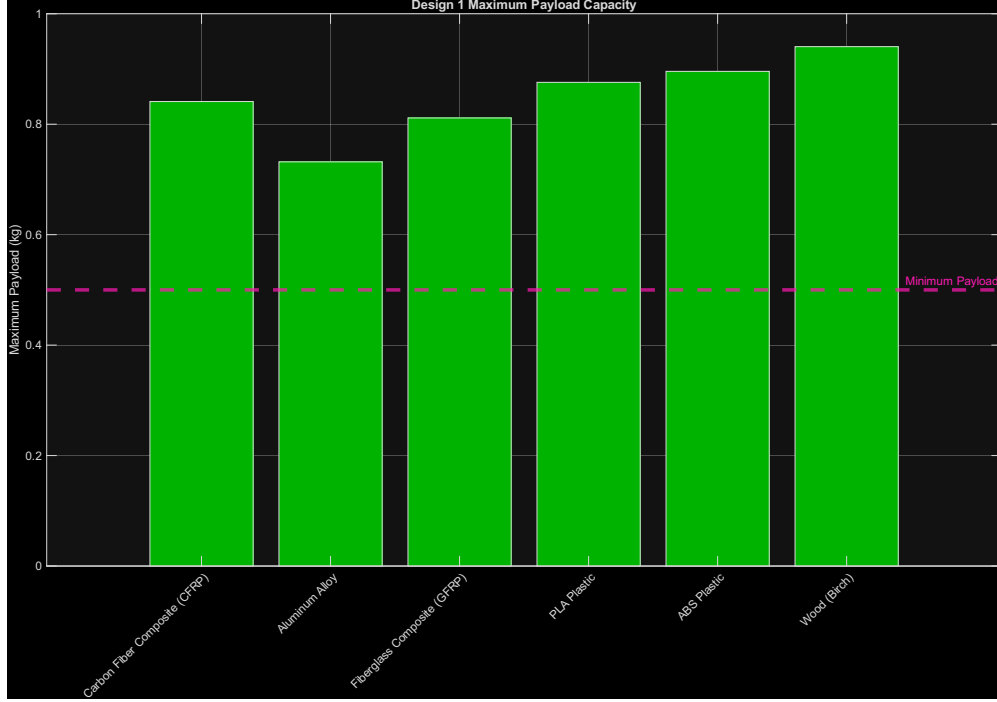


Figure 3: Design 1 Maximum Payload Capacity by Material. All six bars are shown in green, indicating every material passed the minimum payload requirement, marked by the dashed line at 0.5 kg.

3.3 Design 2 Results

Table 6: Design 2 Maximum Payload Results

Material	Total Arm Mass (kg)	Maximum Payload (kg)
Carbon Fiber Composite (CFRP)	0.21949	0.78051
Aluminum Alloy	0.37039	0.62961
Fiberglass Composite (GFRP)	0.26065	0.73935
PLA Plastic	0.17148	0.82852
ABS Plastic	0.14404	0.85596
Wood (Birch)	0.082309	0.91769

Design 2 also passed the payload and thrust to weight requirement for all six materials. No material combination failed.

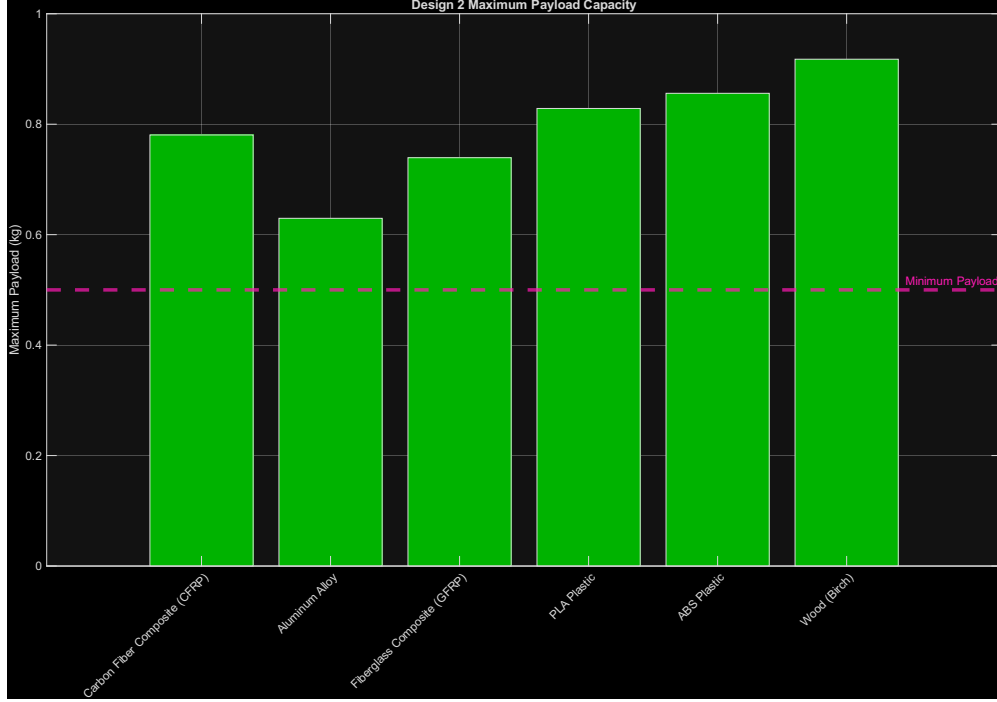


Figure 4: Design 2 Maximum Payload Capacity by Material. All six bars are shown in green, indicating every material passed the minimum payload requirement, marked by the dashed line at 0.5 kg.

3.4 Discussion

Both designs satisfied the payload and thrust to weight requirement across every material option, which means Task 3 alone was not sufficient to separate the two designs from one another. What the results do show clearly is that Design 1 produced a higher maximum payload than Design 2 for every one of the six materials, which follows directly from Design 1's smaller arm volume identified in Section 2.3. For example, using Wood, the lowest mass material tested, Design 1 reached a maximum payload of 0.94040 kg compared to 0.91769 kg for Design 2, and using Aluminum Alloy, the heaviest material tested, Design 1 reached 0.73181 kg compared to 0.62961 kg for Design 2. Since payload capacity alone could not eliminate either design, structural safety and material cost were needed to make a final determination, which is the purpose of Tasks 4 and 5.

4. Finite Element Analysis

Task 4

Finite element analysis was performed on a single drone arm for each design, across all six material options, to evaluate the structural safety factor under motor loading. To simplify the task, this analysis was limited to one drone arm rather than the complete four arm assembly.

4.1 Finite Element Model Setup

Each drone arm geometry was imported into MATLAB using `femodel` with `AnalysisType` set to `structuralStatic`. The Design 1 geometry was imported directly from its STEP file, while the Design 2 geometry, imported from an STL file, required an additional scaling step of 0.001 to convert the model into consistent units. A finite element mesh was then generated for each geometry using `generateMesh` with a maximum element size, `Hmax`, of 1.5×10^{-3} .

Before applying any loads, the geometry was plotted with face labels enabled so the fixed base face and the motor mount face could be visually confirmed against their intended physical locations. This step matters because an incorrectly identified face would silently apply the boundary condition or the load to the wrong location without producing any error, so visually confirming the face IDs against the geometry is a necessary check rather than an optional one.

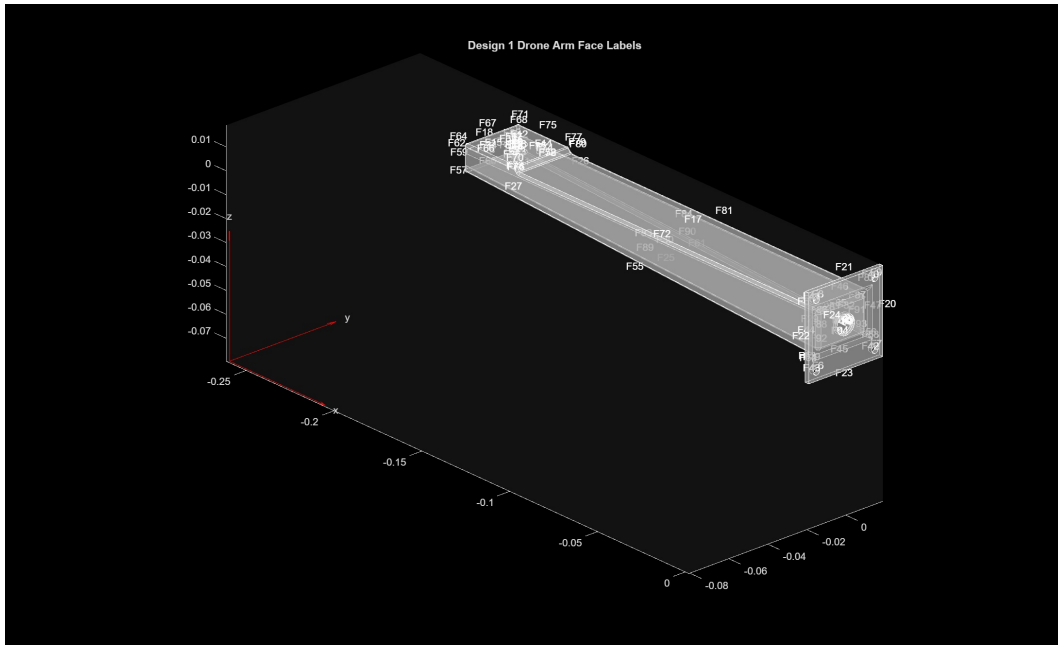


Figure 5: Design 1 Drone Arm Face Labels, Used to Confirm the Fixed Base Face (ID 24) and the Motor Mount Face (ID 33)

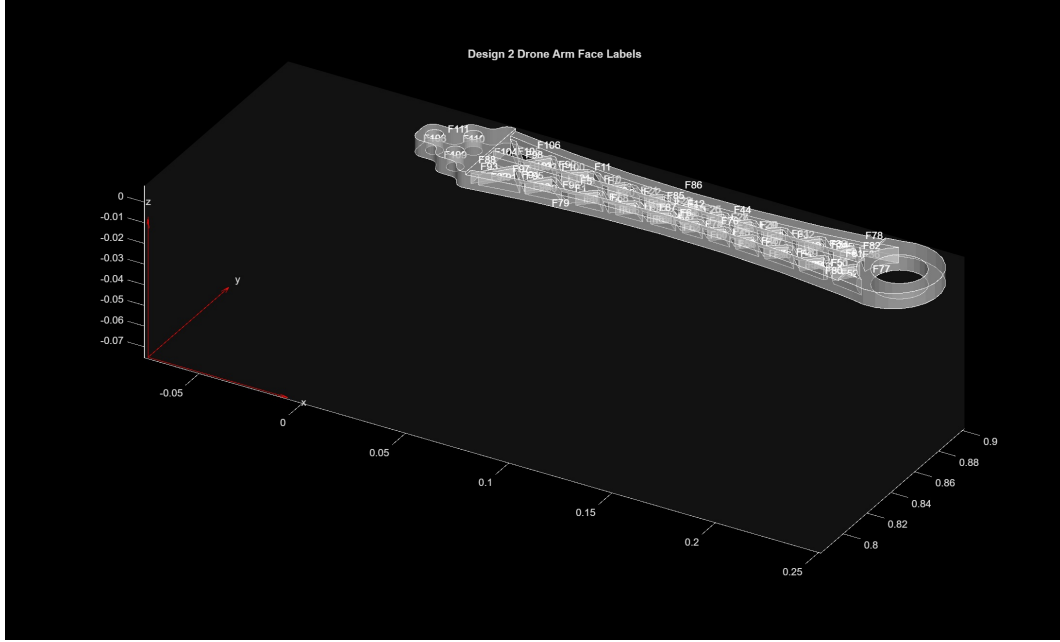


Figure 6: *Design 2 Drone Arm Face Labels, Used to Confirm the Fixed Base Face (ID 111) and the Motor Mount Face (ID 82)*

The fixed base face was constrained using **faceBC** with **Constraint** set to **fixed**, representing the point where the arm attaches to the main drone body. Two loads were then combined into a single net surface traction and applied to the motor mount face: the upward thrust force produced by the motor, calculated as the motor thrust capacity multiplied by gravity, and the downward weight of the motor itself, calculated as the motor mass multiplied by gravity. The net force in the loading direction is the thrust force minus the motor weight force, and that net force was converted into a surface traction by dividing by the motor mount face area, then applied along the positive global Z direction based on each model's coordinate orientation.

For each of the six materials, the model's **MaterialProperties** were set using the Young's modulus and Poisson's ratio stored in **droneArmMaterials.mat**, and the model was solved using **solve**. From each solution, the maximum displacement magnitude was calculated from the combined X, Y, and Z displacement fields, the maximum von Mises stress was extracted directly from the results, and the factor of safety was calculated as the material's yield strength divided by the maximum von Mises stress. A factor of safety at or above 1.5 was treated as meeting the recommended structural safety margin.

4.2 Design 1 Finite Element Results

Table 7: Design 1 Finite Element Analysis Results, All Six Materials

Material	Displacement (mm)	von Mises Stress (MPa)	Factor of Safety
Carbon Fiber Composite (CFRP)	0.059081	4.2484	141.23
Aluminum Alloy	0.059894	4.2349	64.937
Fiberglass Composite (GFRP)	0.1655	4.2546	70.512
PLA Plastic	1.1798	4.2166	14.229
ABS Plastic	2.0652	4.2232	9.4714
Wood (Birch)	0.41304	4.2232	18.943

All six materials passed the finite element safety requirement for Design 1. No material failed.

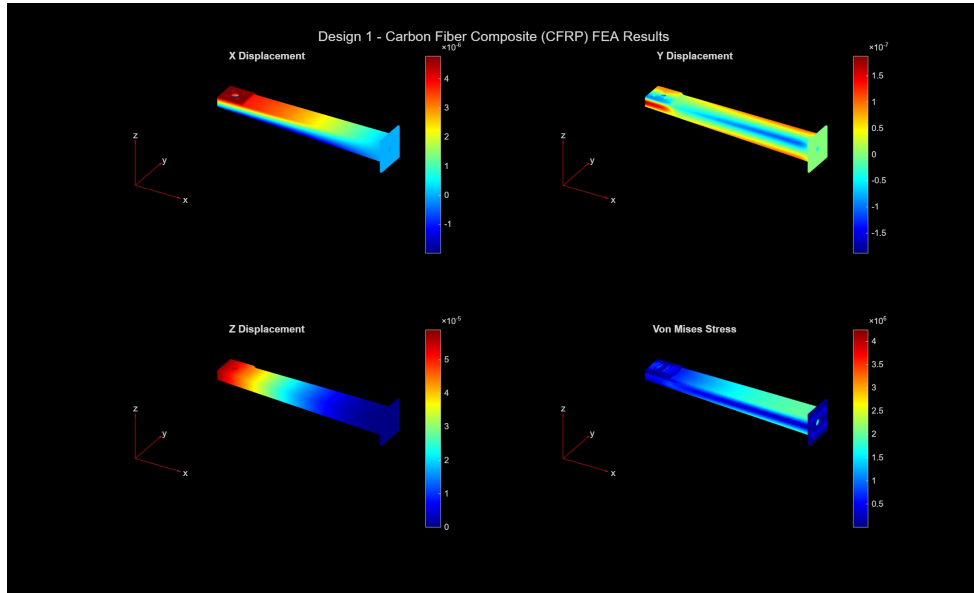


Figure 7: Design 1, Carbon Fiber Composite, X, Y, and Z Displacement and von Mises Stress

Carbon Fiber Composite (CFRP): CFRP produced the lowest maximum displacement of any material tested on Design 1, at 0.059081 mm, and the highest factor of safety of any material tested on the whole project, at 141.23. This is consistent with CFRP having both the highest yield strength, 600 MPa, and a high stiffness among the six materials, which limits deformation under the applied thrust load.

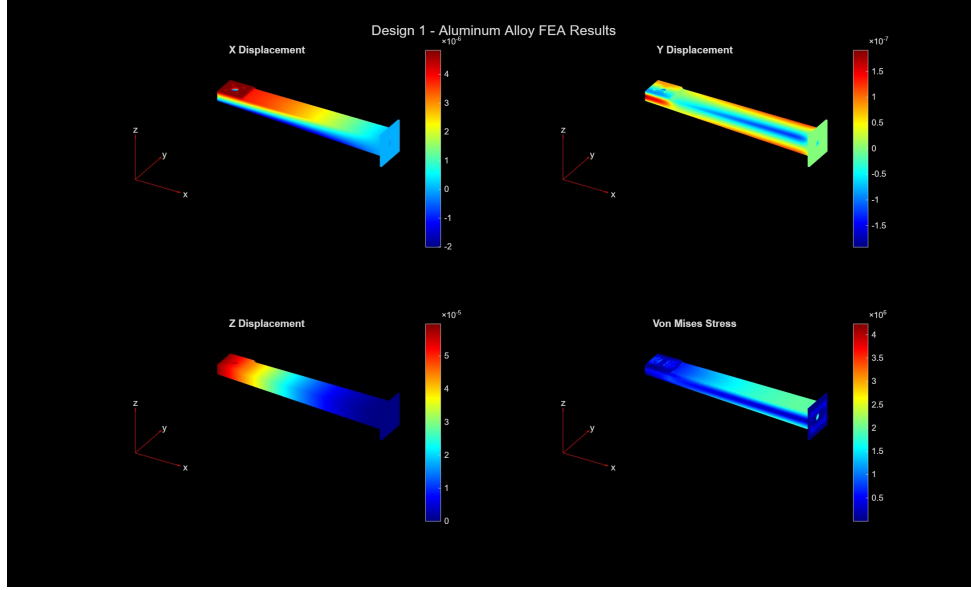


Figure 8: Design 1, Aluminum Alloy, X, Y, and Z Displacement and von Mises Stress

Aluminum Alloy: Aluminum Alloy produced a maximum displacement of 0.059894 mm, nearly identical to CFRP, with a factor of safety of 64.937. Aluminum's lower yield strength relative to CFRP, 275 MPa compared to 600 MPa, is the reason its factor of safety is roughly half of CFRP's despite the two materials showing almost the same displacement.

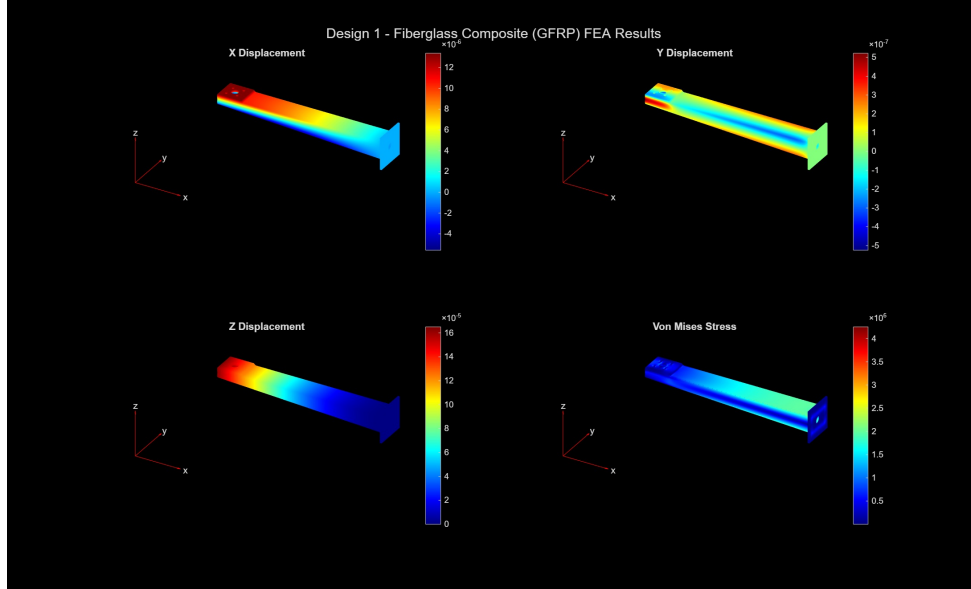


Figure 9: Design 1, Fiberglass Composite, X, Y, and Z Displacement and von Mises Stress

Fiberglass Composite (GFRP): GFRP produced a maximum von Mises stress of 4.2546 MPa, a maximum displacement of 0.1655 mm, and a factor of safety of 70.512. Although GFRP was not the highest performing material on any single metric, it comfortably cleared the 1.5 minimum factor of safety while remaining considerably less expensive than CFRP, which becomes important in Section 5.

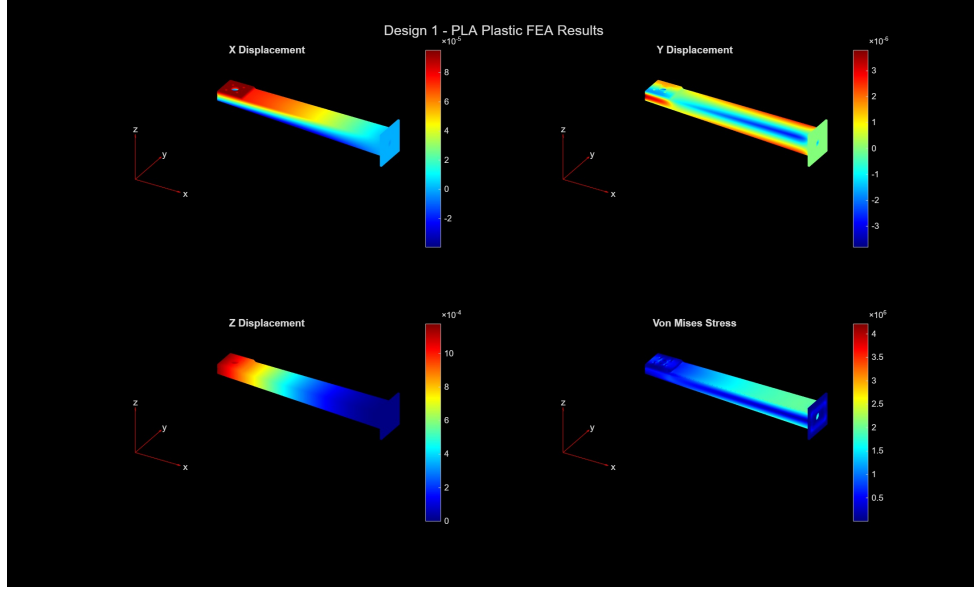


Figure 10: Design 1, PLA Plastic, X, Y, and Z Displacement and von Mises Stress

PLA Plastic: PLA Plastic produced a maximum displacement of 1.1798 mm with a factor of safety of 14.229. While the factor of safety comfortably passes the 1.5 minimum, the displacement value is large enough to raise a separate concern discussed further in Section 6.3, since the tapered section of Design 1 narrows to a wall thickness of only 1.0 mm near the motor mount.

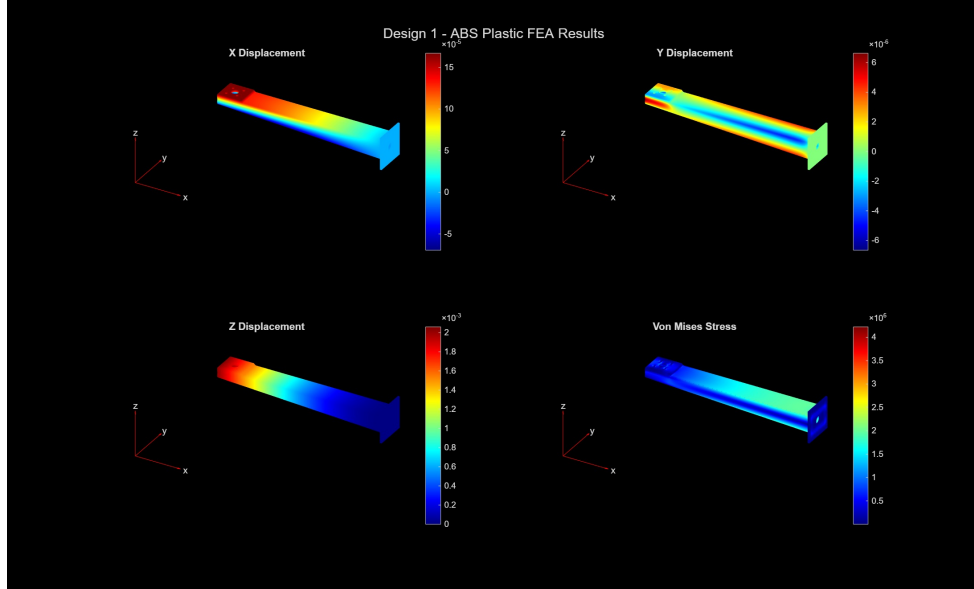


Figure 11: Design 1, ABS Plastic, X, Y, and Z Displacement and von Mises Stress

ABS Plastic: ABS Plastic produced the largest maximum displacement recorded for Design 1, 2.0652 mm, alongside the lowest factor of safety among the materials that still passed, 9.4714. Even though 9.4714 is numerically well above the 1.5 minimum, this combination of a low yield strength material, 40 MPa, with the largest deflection on the design flags ABS as a material that technically passes but is not necessarily the most practical choice, a point returned to directly in Section 6.3.

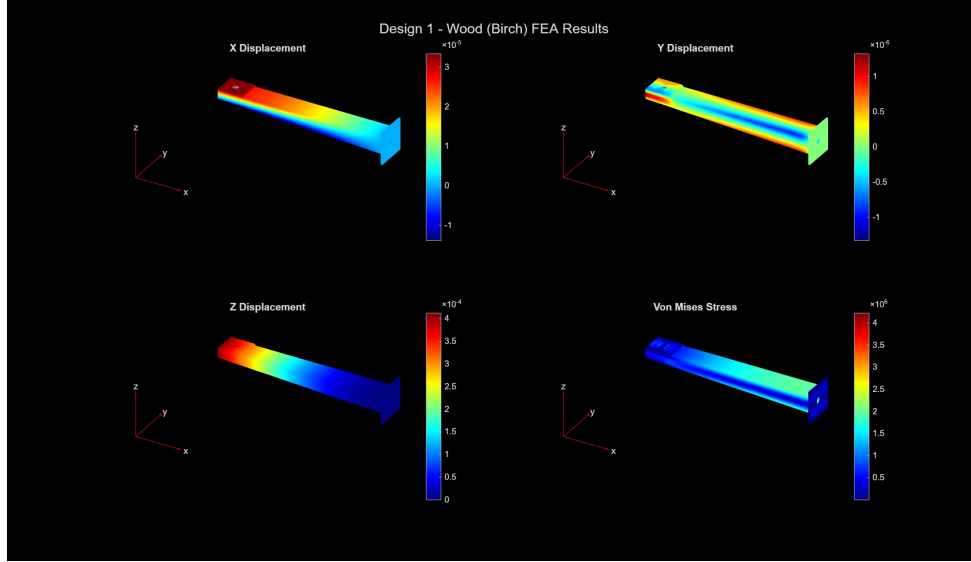


Figure 12: Design 1, Wood (Birch), X, Y, and Z Displacement and von Mises Stress

Wood (Birch): Wood produced a maximum displacement of 0.41304 mm and a factor of safety of 18.943. Wood also produced the highest maximum payload of any material tested on Design 1, 0.94040 kg, and the lowest material cost, which is examined in detail in Sections 5 and 6.

4.3 Design 2 Finite Element Results

Table 8: Design 2 Finite Element Analysis Results, All Six Materials

Material	Displacement (mm)	von Mises Stress (MPa)	Factor of Safety
Carbon Fiber Composite (CFRP)	1.1464	34.414	17.435
Aluminum Alloy	1.1621	34.31	8.0151
Fiberglass Composite (GFRP)	3.2112	34.501	8.6954
PLA Plastic	22.889	34.244	1.7522
ABS Plastic	40.068	34.261	1.1675
Wood (Birch)	8.0137	34.261	2.335

Five of the six materials passed the finite element safety requirement for Design 2. ABS Plastic failed, with a factor of safety of 1.1675, below the 1.5 minimum.

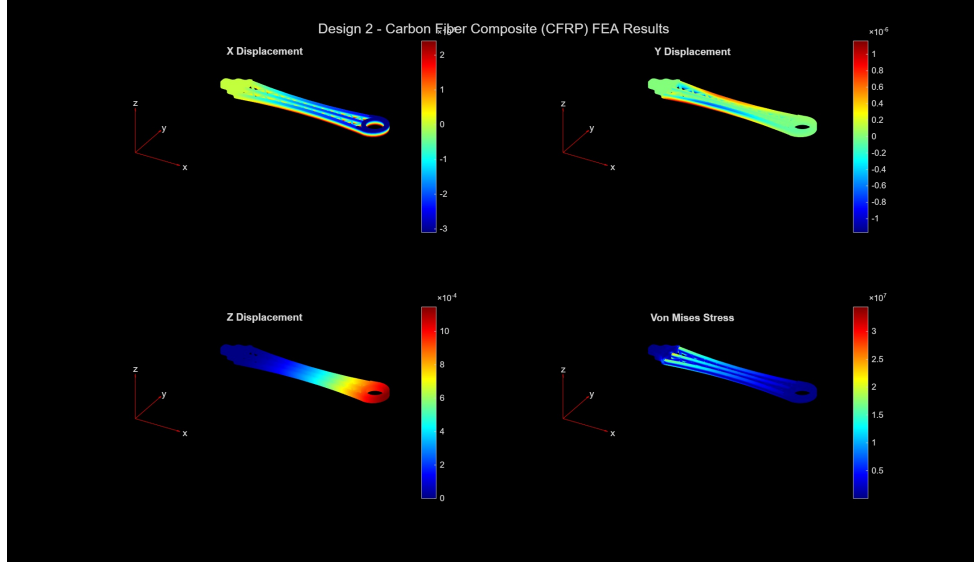


Figure 13: Design 2, Carbon Fiber Composite, X, Y, and Z Displacement and von Mises Stress

Carbon Fiber Composite (CFRP): CFRP produced the lowest maximum displacement on Design 2, 1.1464 mm, and the highest factor of safety on Design 2, 17.435. Even the best performing material and design combination on Design 2 still shows a displacement roughly nineteen times larger than the same material on Design 1, which is the first clear sign that the two geometries do not behave the same way under identical loading.

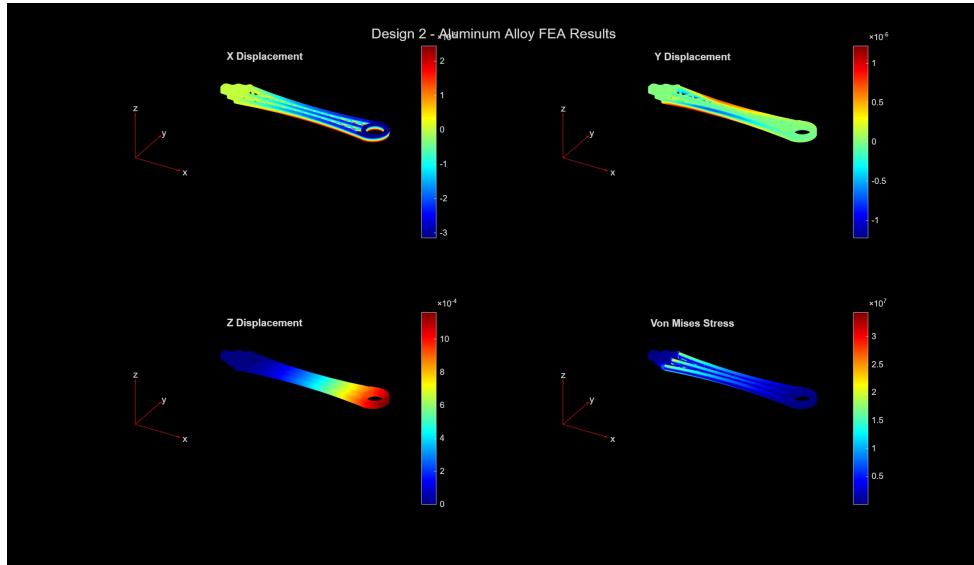


Figure 14: Design 2, Aluminum Alloy, X, Y, and Z Displacement and von Mises Stress

Aluminum Alloy: Aluminum Alloy produced a maximum displacement of 1.1621 mm and a factor of safety of 8.0151 on Design 2, both noticeably weaker than the same material's performance on Design 1.

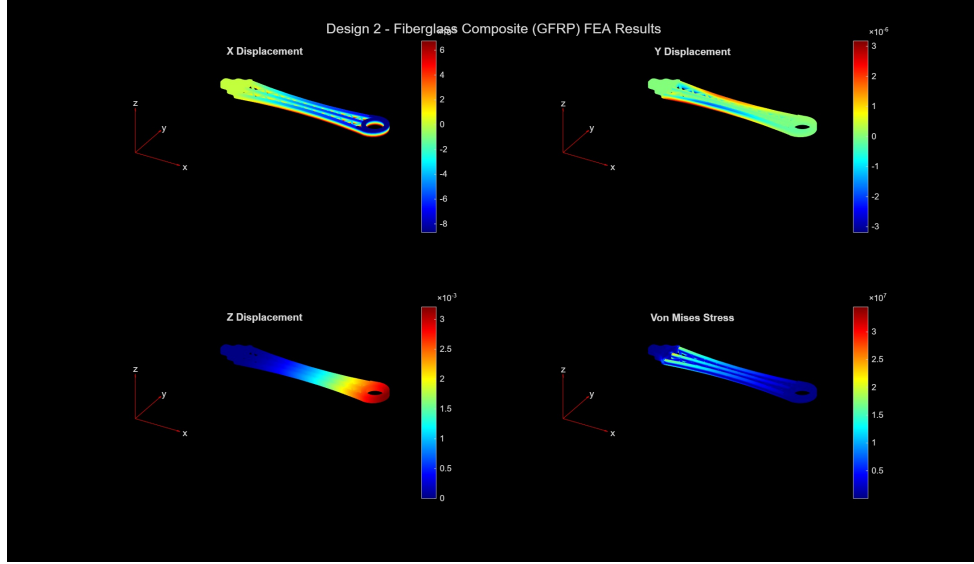


Figure 15: Design 2, Fiberglass Composite, X, Y, and Z Displacement and von Mises Stress

Fiberglass Composite (GFRP): GFRP produced a maximum displacement of 3.2112 mm and a factor of safety of 8.6954 on Design 2. This displacement is roughly nineteen times larger than the 0.1655 mm GFRP produced on Design 1, using the same material properties and a similar loading condition, which reinforces that the difference in behavior comes from geometry rather than material.

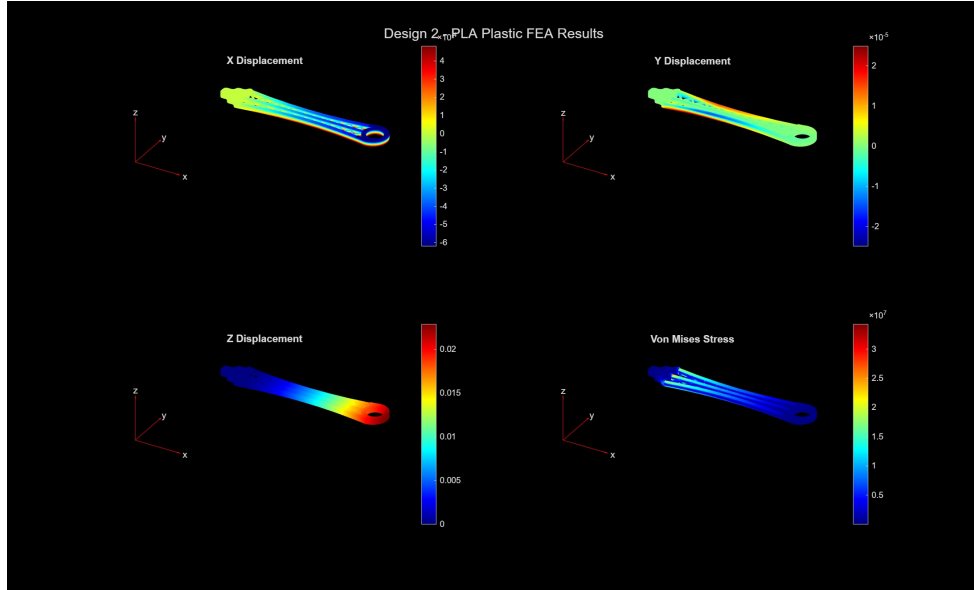


Figure 16: Design 2, PLA Plastic, X, Y, and Z Displacement and von Mises Stress

PLA Plastic: PLA Plastic produced a maximum displacement of 22.889 mm on Design 2, with a factor of safety of 1.7522. Although this factor of safety technically clears the 1.5 minimum, a displacement approaching 23 mm on an arm whose overall height is measured in single digit millimeters represents a structurally unrealistic amount of deflection.

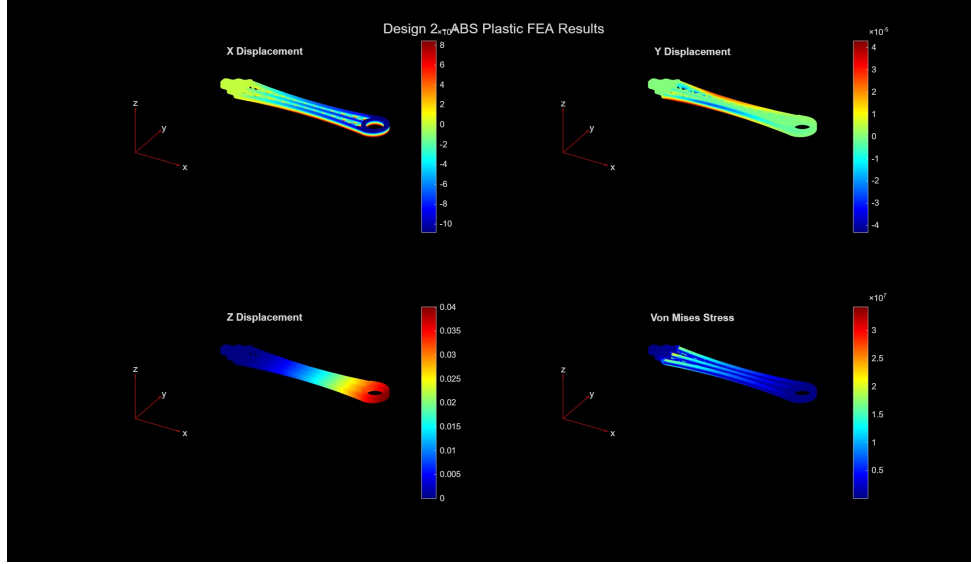


Figure 17: Design 2, ABS Plastic, X, Y, and Z Displacement and von Mises Stress

ABS Plastic: ABS Plastic is the only material and design combination in the entire project that failed the structural safety requirement outright. It produced a maximum displacement of 40.068 mm, by far the largest of any combination tested, with a factor of safety of 1.1675, below the 1.5 minimum target. This combination was excluded from any further consideration in Section 5 and Section 6.

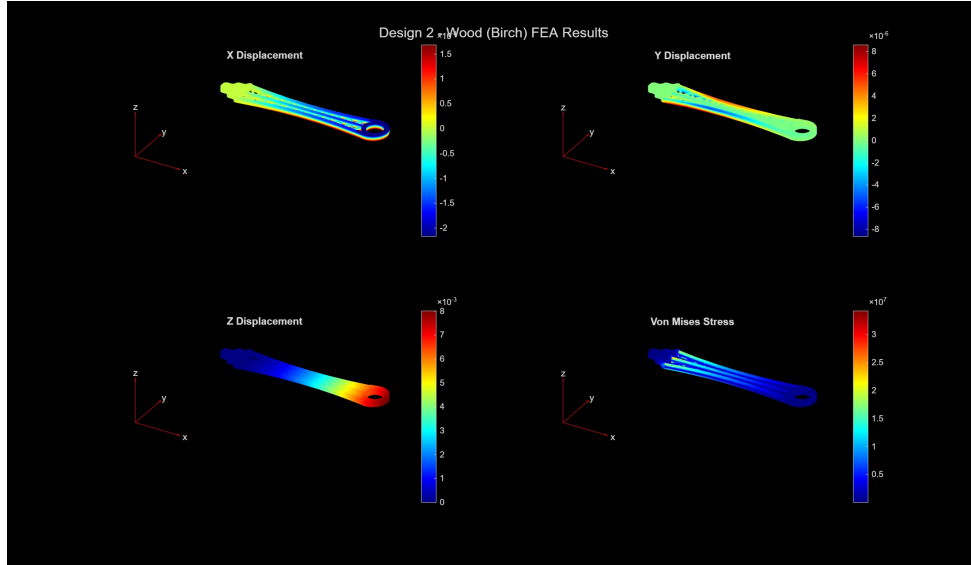


Figure 18: Design 2, Wood (Birch), X, Y, and Z Displacement and von Mises Stress

Wood (Birch): Wood produced a maximum displacement of 8.0137 mm on Design 2, with a factor of safety of 2.335, the lowest passing factor of safety recorded for Design 2. Even the lightest, cheapest material struggled to keep displacement under control once placed on the Design 2 geometry.

4.4 Structural Comparison, Design 1 versus Design 2

Across every one of the six materials, Design 1 produced a smaller maximum displacement and a higher factor of safety than Design 2. The gap is not small. Comparing CFRP on both designs, the strongest material and design combination overall, Design 1 produced a displacement of 0.059081 mm while Design 2 produced 1.1464 mm under the same material and the same load, a difference of roughly nineteen times. That same ratio holds consistently across all six materials, which rules out the possibility that any single material was simply a poor match for the truss geometry. The weakness is coming from the geometry itself.

Before the finite element analysis was run, it was hypothesized in Section 2.2 that the transition region near the arm connection point could be a structural concern for Design 2. The finite element results confirm that hypothesis, and add a more specific explanation for why it occurs. The truss members inside Design 2 are primarily oriented to reinforce the arm in the X and Y directions, which would resist side to side bending and torsion well, but they provide comparatively little resistance in the Z direction, which is the direction the dominant motor thrust and motor weight loads actually act in. Because the truss geometry was not oriented to resist the load direction that matters most for this application, the arm ends up far more flexible under vertical bending than Design 1's continuous tapered wall, which was shaped specifically around that same vertical bending load from the start.

5. Material Cost Optimization

Task 5

Using the results from the thrust to weight analysis and the finite element analysis, this stage determines which material combination provides the lowest material cost while still meeting the required payload, thrust to weight ratio, and factor of safety.

5.1 Methodology

The total material cost for each material option was calculated using $\text{TotalMaterialCost} = \text{CostPerMeter} \times \text{ArmLength} \times \text{NumberOfArms}$, rounded to the nearest cent, using an arm length of 0.202 m for Design 1 and 0.251 m for Design 2. A material and design combination was only considered a valid option if it had already passed both the Task 3 payload and thrust to weight check and the Task 4 factor of safety check. Among the valid options, the material with the lowest total cost was identified as the numerically optimal choice.

5.2 Design 1 Cost Results

Table 9: Design 1 Material Cost Optimization Results

Material	Total Material Cost (USD)	Maximum Payload (kg)	Factor of Safety
Carbon Fiber Composite (CFRP)	24.24	0.84107	141.23
Aluminum Alloy	6.06	0.73181	64.937
Fiberglass Composite (GFRP)	12.12	0.81128	70.512
PLA Plastic	2.83	0.87584	14.229
ABS Plastic	2.83	0.89571	9.4714
Wood (Birch)	1.62	0.94040	18.943

Lowest cost material satisfying all requirements for Design 1: Wood (Birch), at a total material cost of 1.62 USD.

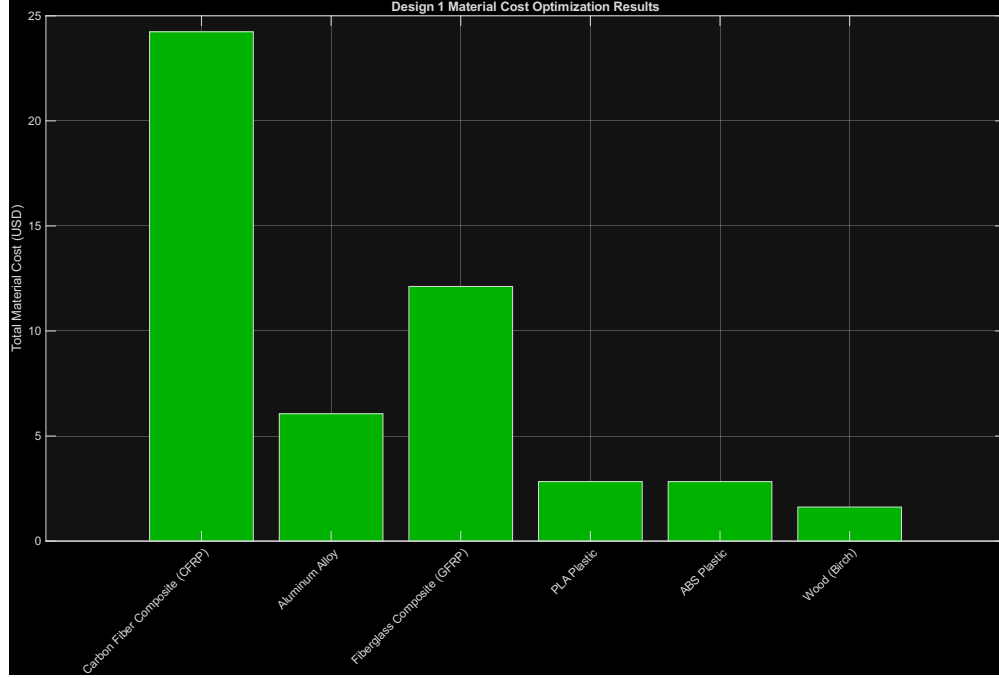


Figure 19: Design 1 Total Material Cost by Material. All six bars are green because all six materials passed both the payload and factor of safety requirements.

5.3 Design 2 Cost Results

Table 10: Design 2 Material Cost Optimization Results

Material	Total Material Cost (USD)	Maximum Payload (kg)	Factor of Safety
Carbon Fiber Composite (CFRP)	30.12	0.78051	17.435
Aluminum Alloy	7.53	0.62961	8.0151
Fiberglass Composite (GFRP)	15.06	0.73935	8.6954
PLA Plastic	3.51	0.82852	1.7522
ABS Plastic	3.51	0.85596	1.1675
Wood (Birch)	2.01	0.91769	2.335

Lowest cost material satisfying all requirements for Design 2: Wood (Birch), at a total material cost of 2.01 USD. ABS Plastic is excluded from consideration here because it failed the Task 4 factor of safety requirement.

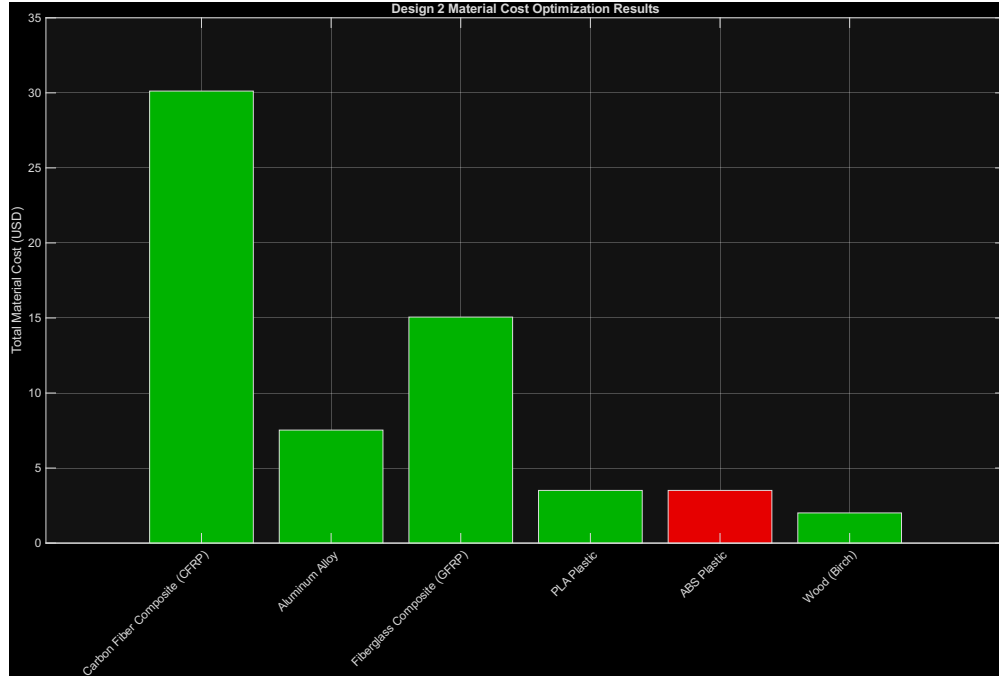


Figure 20: Design 2 Total Material Cost by Material. Five bars are green, indicating those materials passed both the payload and factor of safety requirements. ABS Plastic is shown in red because it failed the factor of safety requirement in Task 4.

5.4 Discussion

Wood was the lowest cost material satisfying every requirement on both designs, at 1.62 USD on Design 1 and 2.01 USD on Design 2, and it also produced the highest maximum payload of any material on both designs. On cost and payload numbers alone, Wood looks like the strongest possible choice on either geometry. However, cost and payload capacity are only two of the factors that go into a practical material recommendation, and the reasoning behind why Wood was ultimately not selected as the final recommendation is addressed directly in Section 6.4.

Comparing the two designs directly, every material costs more on Design 2 than the equivalent material on Design 1, which follows directly from Design 2's larger arm volume identified back in Section 2.3. Combined with the structural comparison in Section 4.4, Design 2 is both more expensive and less structurally sound than Design 1 across every material that was tested on both geometries.

6. Final Design Recommendation and Interpretation of Results

Interpretation of Results, Final Recommended Design Solution

Using the results of the thrust to weight analysis, the finite element analysis, and the material cost optimization together, this section identifies the final recommended drone arm design and material, explains the tradeoffs that were weighed to reach that recommendation, and compares the selected design against the alternative that was not chosen.

6.1 Selected Design and Material

The final recommended drone arm design is Design 1, the hollow tapered rectangular tube. The final recommended material for this design is Fiberglass Composite (GFRP). This recommendation was based not only on the quantitative results, including maximum payload capacity, maximum displacement, factor of safety, and cost, but also on qualitative feasibility such as manufacturability and real world material behavior that cannot be fully captured through this MATLAB analysis alone.

Before weighing the individual requirements, it is worth restating the geometric context established in Section 2. Design 1 strategically allocates material where it is most needed. Since the arm experiences the highest bending loads near the connection point to the main drone body, Design 1 was designed with larger cross sectional dimensions and thicker walls near the base, while reducing material toward the motor mount, where less structural reinforcement is required. This allowed Design 1 to maintain the necessary structural capability while minimizing unnecessary material usage. Design 2, by comparison, uses a hollow truss shaped internal structure intended to increase rigidity while reducing mass, but its truss members are oriented in a way that made them less effective at resisting the dominant vertical bending loads generated by motor thrust, a limitation that was hypothesized before the analysis and confirmed by the finite element results in Section 4.4.

6.2 Payload Capacity Justification

The first constraint is that the design must support a minimum payload of 0.5 kg and achieve a thrust to weight ratio of at least 2 to 1. Given that the quadcopter has four motors, each producing 1 kg of thrust, the maximum allowable weight of the entire quadcopter is 2 kg. Since 1 kg is reserved for components other than the drone arms, this leaves a weight budget of 1 kg for the four drone arms and the minimum payload combined. Based on the design sketches and the resulting CAD volumes, both designs passed this requirement across all six materials. More importantly, the findings indicated that Design 1 had a smaller volume than Design 2, approximately $2.483 \times 10^{-5} \text{ m}^3$ (24.83 cm³) compared to $3.430 \times 10^{-5} \text{ m}^3$ (34.30 cm³), and therefore less overall weight, which gave an early indication that Design 1 was the stronger option even before the payload numbers were calculated.

As for the materials, GFRP used on Design 1 allowed for a maximum payload capacity of 0.81128 kg. This was not the highest performing material on this metric alone, since CFRP, PLA, ABS, and Wood all produced higher maximum payload capacities of 0.84107 kg, 0.87584 kg, 0.89571 kg, and 0.94040 kg respectively. Only Aluminum Alloy performed worse on payload capacity, producing 0.73181 kg. Payload capacity by itself was therefore not the deciding factor in selecting GFRP, since four of the six materials outperformed it on this measure alone.

6.3 Structural Safety Justification

The second requirement is that the engineering safety margin, represented by the factor of safety, must be approximately 1.5 to 2 or higher under the given loading conditions, with the motor mount face loaded and the root attachment flange fixed to represent the connection point to the main drone body. Although this is a simplified representation of the actual loading conditions experienced during flight, it still produced meaningful results for comparing the structural performance of each material. All six materials passed the required safety criteria on Design 1. GFRP specifically produced a maximum von Mises stress of 4.2546 MPa, a maximum displacement of 0.1655 mm, and a factor of safety of 70.512.

GFRP was still not the highest performing option on this measure either. CFRP produced the lowest maximum displacement of any material at 0.059081 mm and the highest factor of safety at 141.23, while Aluminum Alloy produced a maximum displacement of 0.059894 mm and a factor of safety of 64.937. Wood produced a higher maximum displacement of 0.41304 mm and a factor of safety of 18.943. The largest concern appeared with the plastic materials, specifically PLA and ABS. While both materials technically passed the factor of safety requirement, PLA produced a maximum displacement of 1.1798 mm with a factor of safety of 14.229, and ABS produced a maximum displacement of 2.0652 mm with a factor of safety of 9.4714.

These displacement values may seem acceptable on their own, but when compared against the actual wall thickness of Design 1, which tapers down to approximately 1.0 mm near the motor mount, the concern becomes clearer. The motor thrust cannot be assumed to remain perfectly aligned with the vertical loading direction once the displacement grows larger than roughly twice the thickness of the thinnest section of the arm. In real world conditions, this magnitude of deformation would introduce vibration and dynamic instability, an effect referred to in drone applications as the jello effect. Although PLA and ABS satisfy the numerical factor of safety requirement on paper, their excessive deformation makes them a poor practical material choice for this specific application, which is part of the reason GFRP, with its much smaller 0.1655 mm displacement, was preferred over materials that technically passed but deformed far more under the same load.

6.4 Material Cost Justification

Finally, a cost optimization was performed to determine which material satisfied all numerical requirements while minimizing total material cost. Although GFRP resulted in a total material cost of 12.12 USD on Design 1, it was not the least expensive option. Wood produced the lowest material cost at only 1.62 USD, followed by PLA and ABS at 2.83 USD each, Aluminum Alloy at 6.06 USD, GFRP at 12.12 USD, and CFRP at 24.24 USD.

Based on the numerical optimization alone, Wood would appear to be the optimal material choice, since it also achieved the highest maximum payload capacity of 0.94040 kg while maintaining a factor of safety of 18.943. However, this optimization does not capture several important real world engineering considerations. Since Wood is a natural material, its mechanical properties are not perfectly consistent, due to grain orientation, knots, moisture absorption, and manufacturing variability. Despite its strong numerical performance and very low cost, Wood was therefore not considered the most practical material for a drone arm intended for repeated real world flight. At the other end of the spectrum, CFRP provided the highest structural performance overall but was ultimately judged unnecessary once GFRP was shown to produce similarly strong results for close to half the cost. GFRP was therefore selected as the material that best balances structural performance, manufacturing practicality, material consistency, and overall cost, rather than the material that wins outright on any single metric.

6.5 Comparison to Design 2

Compared with Design 2, Design 1 demonstrated meaningfully better structural behavior across every material tested. Although Design 2 technically met the project’s minimum requirements for five of the six materials, it consistently experienced far greater maximum displacement than Design 1 under the identical loading condition, as shown directly in Section 4.4. Through investigation, it was determined that the truss members inside Design 2 primarily reinforced the X and Y directions but provided limited resistance in the Z direction, which reduced the arm’s resistance to vertical bending and was the likely contributor to its consistently higher displacement across all six materials.

Rather than redesigning Design 2’s truss orientation after seeing these results, the team chose to retain the original design as tested, because it demonstrates the value of the engineering design process itself. The results highlight how analysis, testing, and iteration lead to better informed engineering decisions, and they provide a useful lesson for any future revision of this arm: a truss structure is only as effective as the direction its members are oriented relative to the load it actually needs to resist.

6.6 Summary Recommendation

Overall, Design 1 manufactured from GFRP provided the best combination of payload capacity, structural integrity, manufacturability, and material cost of any design and material combination tested in this project. It was not the top performer on any single metric in isolation. CFRP was stronger, Wood was cheaper and lighter, and several materials produced a higher raw payload number. What made GFRP on Design 1 the recommended choice is that it cleared every requirement comfortably while avoiding the specific weaknesses identified in the alternatives, excessive deformation in the plastics, questionable long term consistency in Wood, and unnecessary cost in CFRP, all on top of the geometry that already proved structurally superior to Design 2 in Section 4.4.

7. Limitations and Future Work

Several limitations should be considered when interpreting these results.

- The accuracy of the finite element analysis depends on mesh size. A reasonably refined mesh that demonstrated satisfactory convergence of the results was used for this project, although a more comprehensive mesh convergence study with further refinement should be performed in future work to confirm the reported values do not shift meaningfully with a finer mesh.
- The finite element analysis only evaluated a single drone arm rather than the complete four arm drone assembly, meaning interactions between the arms, the central frame, fasteners, and the motor mounting hardware were not included in this analysis.
- The material models assumed isotropic behavior, which simplifies the actual behavior of composite materials such as GFRP and CFRP, whose real properties depend on fiber orientation and manufacturing process, and also simplifies Wood, whose properties depend on grain direction.
- The applied loading conditions were simplified by representing the motor thrust and motor weight as a single combined surface traction, without accounting for real flight effects such as vibration, acceleration, impact loading, aerodynamic forces, or changing thrust conditions during maneuvers.
- The calculated factor of safety does not account for fatigue from repeated flight cycles, manufacturing defects, environmental effects such as temperature or humidity, or long term material degradation.

For future work, the drone assembly could be tested using more realistic loading conditions and material properties, including motor torque and off axis thrust effects, while validating the numerical results through physical prototype testing. Incorporating wind loading, vibration analysis, landing impact loads, and a mesh convergence study would further strengthen confidence in the recommended design before committing to a physical build.

8. Team Roles and Project Management

This project was completed by a four person team under a written teamwork agreement that defined roles, communication norms, and quality standards before analysis work began.

Table 11: Team Member Roles

Team Member	Role
Julianna Amaya	Quality Assurance and Reproducibility Lead, Modeling Lead
Boula (Paul) Etans	Modeling Lead, Documentation and Visualization Lead
Mario Mendoza	Analysis and Validation Lead
Nathan Castro	Project Manager

The Project Manager scheduled meetings, maintained the project timeline, and ensured tasks were tracked and disagreements were resolved. The Modeling Lead owned the core MATLAB models and simulations. The Analysis and Validation Lead performed hand calculations and cross checked simulation outputs against expected physical behavior. The Documentation and Visualization Lead compiled the report, figures, and tables, and built the visual artifacts included throughout this document. The Quality Assurance and Reproducibility Lead enforced a coherent style across all project artifacts, made sure work was not overwritten across different versions, and re-ran code to confirm robustness and reproducibility.

The team communicated primarily through Discord, with weekly meetings held on Mondays for 60 to 90 minutes, and a response time standard of within 24 hours on weekdays and within 48 hours on weekends. Version control was maintained through a single shared GitHub repository, with decision records kept in a shared Google Drive. The team’s definition of done required that all requirements met the project description, that the code was reproducible directly from the README file, that all specified figures could be regenerated, that at least one teammate had tested edge cases, and that all assumptions, units, and references were documented in the final report.

The project was completed against a four week internal timeline. Week 1 focused on completing Task 1 and sketching both proposed designs. Week 2 focused on finishing Task 2 by building CAD models of both sketches and completing Task 3 by confirming both designs passed the project’s payload and thrust to weight requirements. Week 3 focused on finishing Task 4 by running finite element analysis on both designs across all six required materials and determining the design and material combination with the best overall structural integrity and performance. Week 4 focused on completing Task 5 by identifying the lowest cost material satisfying all requirements for both designs, and on finalizing documentation for submission to the shared GitHub repository.

Conclusion

This project developed a MATLAB based workflow that evaluates drone arm geometry and material selection together rather than in isolation, using thrust to weight analysis, single arm finite element analysis, and material cost optimization across two structurally distinct designs and six candidate materials. Twelve design and material combinations were evaluated in total, and eleven of the twelve satisfied every project requirement, with only Design 2 manufactured from ABS Plastic failing the structural safety check.

The analysis showed that geometry mattered more than any single material choice. Across every one of the six materials tested, Design 1 outperformed Design 2 on displacement, factor of safety, payload capacity, and cost, a result traced back to the orientation of Design 2's internal truss members relative to the dominant vertical bending load rather than to any weakness in the materials themselves. Within Design 1, no single material dominated every metric. CFRP was the strongest, Wood was the cheapest and lightest, and several plastics passed the numerical safety check while deforming enough to raise practical concerns about vibration and flight stability. Weighing payload capacity, structural safety, real world material behavior, and cost together, Design 1 manufactured from Fiberglass Composite was selected as the design that best satisfies the full set of project requirements without relying on being the top performer in any single category.

We hope this project demonstrates how a structured, quantitative MATLAB workflow, comparing multiple geometries against multiple materials under the same conditions, can support a real engineering design decision with evidence rather than assumption, and how the same workflow can be extended in future work to more realistic loading conditions and physical prototype validation.

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